

RBG-48371

10 CFR 50.73

August 14, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2025-003-00, Turbine Control Valve
Closure Results in Loss of Safety Function

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Jack McCoy, Regulatory and Emergency Planning Manager, at 225-378-3310.

Respectfully,

**Jack
McCoy** 
Digitally signed by Jack McCoy
DN: cn=Jack McCoy, o=Manager
Regulatory and Emergency Preparedness,
ou=Entergy/ River Bend,
email=jmccoy1@entergy.com
Date: 2025.08.14 09:15:39 -05'00'

JM/jdb

Enclosure: LER 50-458 / 2025-003-00, Turbine Control Valve
Closure Results in Loss of Safety Function

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

RBG-48371

Licensee Event Report 50-
458/2025-003-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name River Bend Station, Unit 1	☒ 050	2. Docket Number 00458	3. Page 1 of 2
	☐ 052		

4. Title
Turbine Control Valve Closure Results in Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050 Docket Number
06	18	2025	2025	003	00	08	14	2025	N/A	
									Facility Name	☐ 052 Docket Number
									N/A	

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)(i)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Jack McCoy, Regulatory and Emergency Planning Manager	Phone Number (Include Area Code) (225) 378-3310
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
B	TG	IMOD	G080	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 18, 2025, at 0553 CDT, with River Bend Station (RBS), Unit 1 operating in Mode 1 at 100% power, High Pressure Turbine Main Steam Supply Line "D" Control Valve (MSS-HYVCV3) closed causing both Turbine Bypass Valves to open. Reactor power was lowered to approximately 95% and the Bypass Valves Closed. At 0646 CDT, MSS-HYVCV3 was observed opening and all other Turbine Control Valves intermittently changing position causing the Bypass Valves to open again. Reactor power was lowered to approximately 90% and the Bypass Valves closed. The Bypass Valves were open for approximately one minute for each of the two occurrences.

With the Bypass Valves open, Technical Specification (TS) required instrumentation that utilize turbine first stage pressure were declared inoperable.

The direct cause of this event was a failure of the Redundant Valve Positioner "B" (RVP-B) for MSS-HYVCV3. The RVP-B was replaced on June 19, 2025.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(A) and 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of the safety function of structures or systems that are needed to: (A) shut down the reactor and maintain it in a safe shutdown condition and (D) mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station, Unit 1	☒ 050	00458	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	003	00

NARRATIVE**Event Description**

On June 18, 2025, at 0553 CDT, with River Bend Station (RBS), Unit 1 operating in Mode 1 at 100% power, the High-Pressure Turbine Main Steam Supply Line "D" Control Valve (MSS-HYVCV3) [TG:VLV] closed causing both Turbine Bypass Valves [JI:V] to open. Control room operators entered appropriate procedures and began to lower reactor power. Reactor power was lowered to approximately 95% and the Turbine Bypass Valves closed. At 0646 CDT, MSS-HYVCV3 was observed opening and all other High-Pressure Turbine Main Steam Supply Line Control Valves were observed intermittently changing positions causing the Turbine Bypass Valves to open again. Reactor power was lowered to approximately 90% and the Bypass Valves closed.

During the time the Turbine Bypass Valves were open, Technical Specification (TS) required instrumentation that utilizes turbine first stage pressure were declared inoperable. Safety Related Functions of TS 3.3.1.1 (RPS Instrumentation), TS 3.3.2.1 (Control Rod Block Instrumentation) and TS 3.3.4.1 (End of Cycle Recirculation Pump Trip) were declared inoperable.

This event was reported on June 18, 2025, at 1126 CDT in accordance with 10 CFR 50.72(b)(3)(v)(A) and 10 CFR 50.72(b)(3)(v)(D) as an event or condition that at the time of discovery could have prevented fulfillment of a safety function of structures or systems that are needed to: (A) shut down the reactor and maintain it in a safe shutdown condition and (D) mitigate the consequences of an accident. (EN 57767)

Event Cause

The direct cause of this event was a failure of the Redundant Valve Positioner "B" (RVP-B) for MSS-HYVCV3. This failure resulted in the closing of MSS-HYVCV3 resulting in the opening of the Turbine Bypass Valves to maintain reactor pressure at the selected setpoint. The RVP-B was replaced on June 19, 2025.

Safety Assessment

During the transient there were no radiological impacts and therefore no actual impact to the health and safety of the public. The event is considered of small safety significance from the Probabilistic Risk Assessment (PRA) perspective as no PRA related mitigative components or strategies were affected by this event. While the corresponding functions were declared inoperable from a Technical Specification perspective, functionality was maintained, therefore this is not considered a safety system functional failure.

Corrective Actions**Planned:**

- Perform a vendor analysis for the failed RVP module.
- Investigate a possible solution to mitigate or eliminate similar failures of the RVP modules.

Previous Occurrences

None

Energy Industry Identification System (EIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).